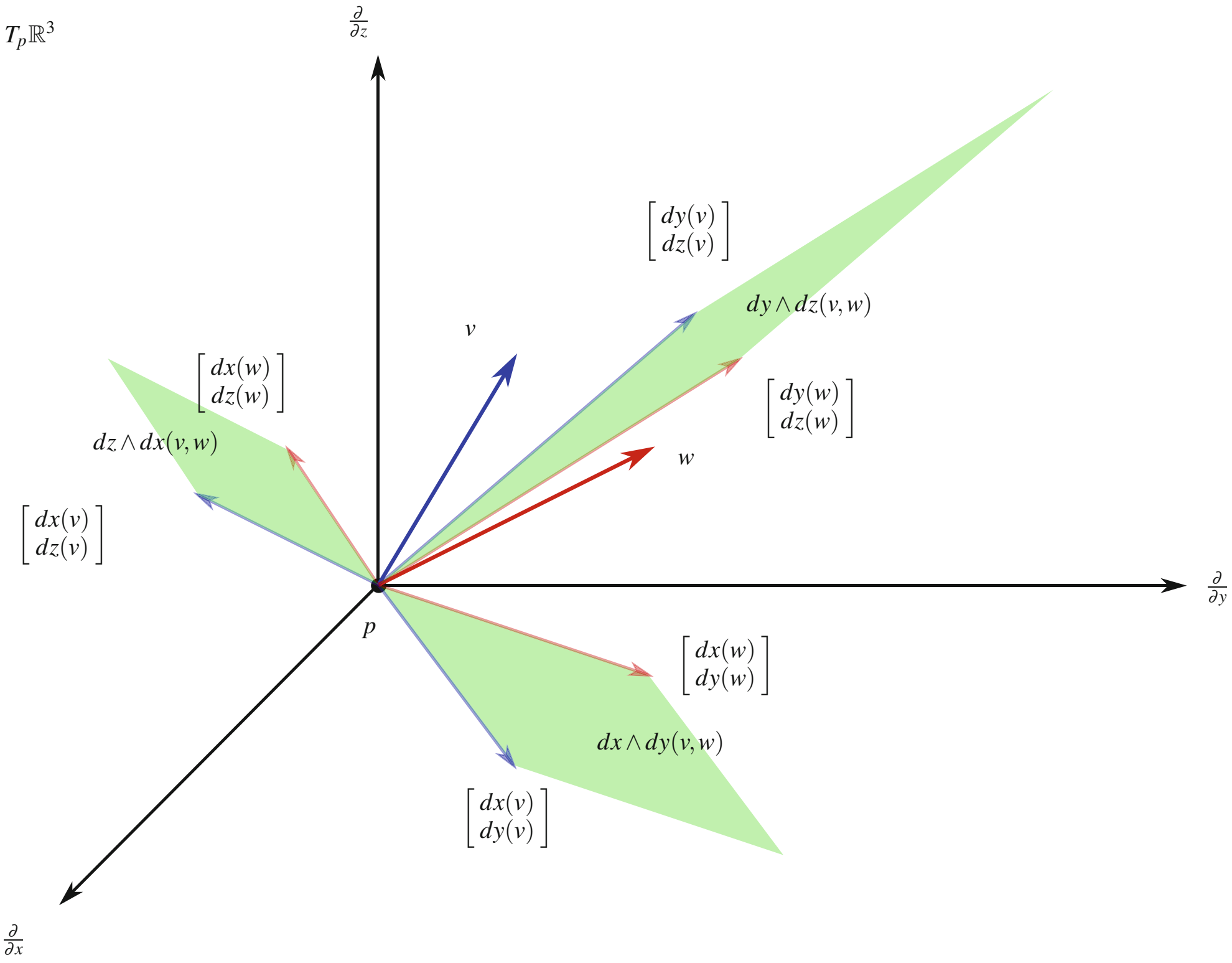

$T_p\mathbb{R}^3$
$\frac{\partial}{\partial z}$
$\begin{bmatrix} dy(v) \\ dz(v) \end{bmatrix}$
$dy \wedge dz(v,w)$
v
$\begin{bmatrix} dx(w) \\ dz(w) \end{bmatrix}$
$\begin{bmatrix} dy(w) \\ dz(w) \end{bmatrix}$
$dz \wedge dx(v,w)$
w
$\begin{bmatrix} dx(v) \\ dz(v) \end{bmatrix}$
$\frac{\partial}{\partial y}$
p
$\begin{bmatrix} dx(w) \\ dy(w) \end{bmatrix}$
$dx \wedge dy(v,w)$
$\begin{bmatrix} dx(v) \\ dy(v) \end{bmatrix}$
$\frac{\partial}{\partial x}$